Telecom Customer Churn Analysis

Executive Summary

This project analyzes telecom customer churn behavior using exploratory data analysis, statistical validation, and revenue impact modeling.

Key Findings:

• Month-to-month contract customers show highest churn risk. • Early tenure customers (0–12 months) contribute majority churn volume. • High monthly charges increase churn probability. • Pareto analysis shows 38.41% customers generate 80% revenue.

Business Recommendation:

Focus retention programs on high-value early tenure customers to reduce churn-driven revenue loss.

Business Problem

A telecom company is experiencing customer churn. Customer churn directly impacts revenue, customer acquisition cost, and long-term profitability.

Objective

Identify:

- Who is churning?
- Why they are churning?
- What factors influence churn?
- Which customers are high-risk?
- What actions can reduce churn?

Business Impact

Reducing churn by even 5% can significantly increase revenue and customer lifetime value.

This project applies structured data analysis, statistical validation, and business interpretation to generate actionable insights.

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("telecom data/Telecom-Customer-Churn.csv")

df.head()
```

Out[1]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	M
0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	

5 rows × 21 columns

DATA UNDERSTANDING

Dataset Columns Overview

- CustomerID — Unique customer identifier.
- Contract — Contract type of customer.
- Tenure — Number of months customer stayed.
- MonthlyCharges — Monthly subscription fee.
- TotalCharges — Total revenue generated.
- Churn — Customer churn status (Yes/No).

In [2]:
```
df.shape
```

Out[2]:
```
(7043, 21)
```

In [3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

In [4]:
```
df['Churn'].value_counts()
```

Out[4]:
```
Churn
No     5174
Yes    1869
Name: count, dtype: int64
```

In [5]:
```
df['Churn'].value_counts(normalize=True)*100
```

Out[5]:
```
Churn
No     73.463013
Yes    26.536987
Name: proportion, dtype: float64
```

Initial Insight

- Total Customers: 7043
- Churn Rate: 26.5%

This means approximately 1 in 4 customers leave the company.

This is a serious revenue leakage problem.

DATA CLEANING

In [6]:
```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'],errors='coerce')
df.isnull().sum()
```

Out[6]:
```
customerID           0
gender               0
SeniorCitizen        0
Partner              0
Dependents           0
tenure               0
PhoneService         0
MultipleLines        0
InternetService      0
OnlineSecurity       0
OnlineBackup         0
DeviceProtection     0
TechSupport          0
StreamingTV          0
StreamingMovies      0
Contract             0
PaperlessBilling     0
PaymentMethod        0
MonthlyCharges       0
TotalCharges        11
Churn                0
dtype: int64
```

In [7]:
```
df = df.dropna()
df.shape
```

Out[7]:
```
(7032, 21)
```

EDA

Question 1-- who churns more by contract type?

In [8]:
```
#  Contract vs Churn
pd.crosstab(df['Contract'],df['Churn'],normalize='index').round(4)*100
```

Out[8]:

Churn	No	Yes
Contract		
Month-to-month	57.29	42.71
One year	88.72	11.28
Two year	97.15	2.85

Business Insight

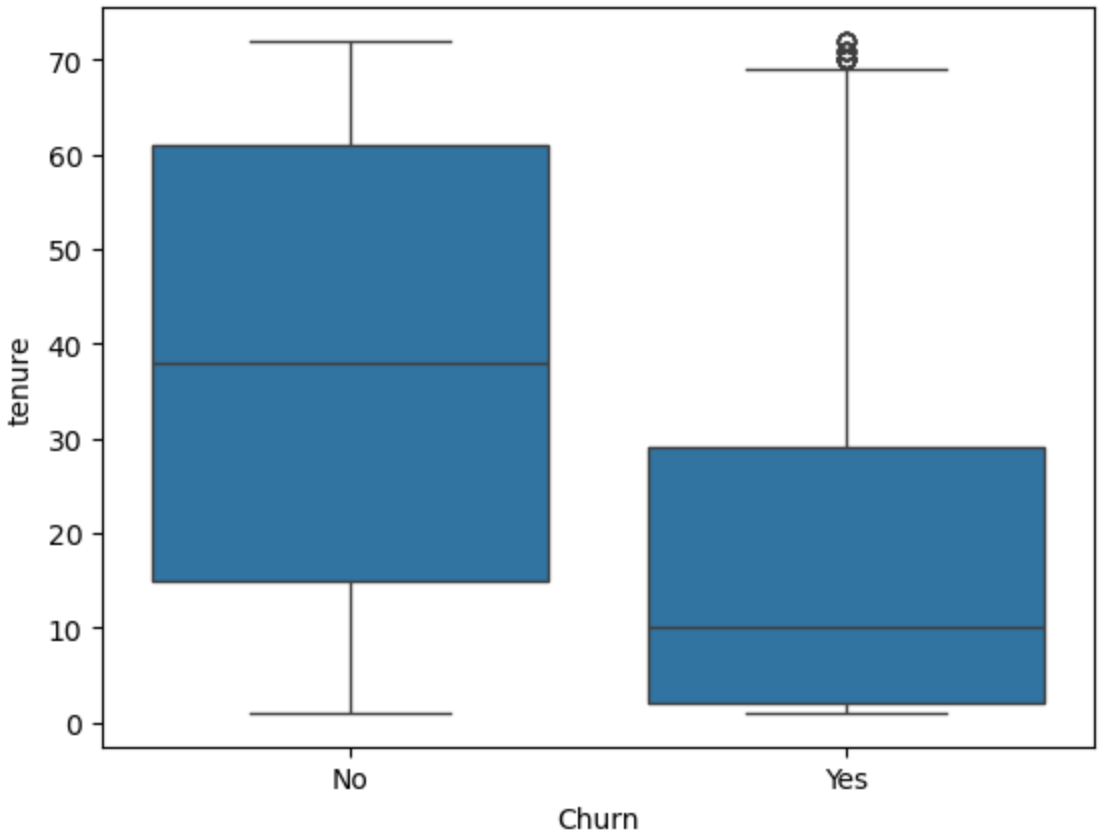
Month-to-month customers are extremely high churn risk.

Long-term contracts reduce churn drastically.

- Action: Incentivize longer contracts.

Question 2: Tenure vs Churn

In [9]:
```
sns.boxplot(x='Churn',y='tenure',data=df)
plt.show()
```



- Churn customers have very low tenure
- loyal customers have high tenure

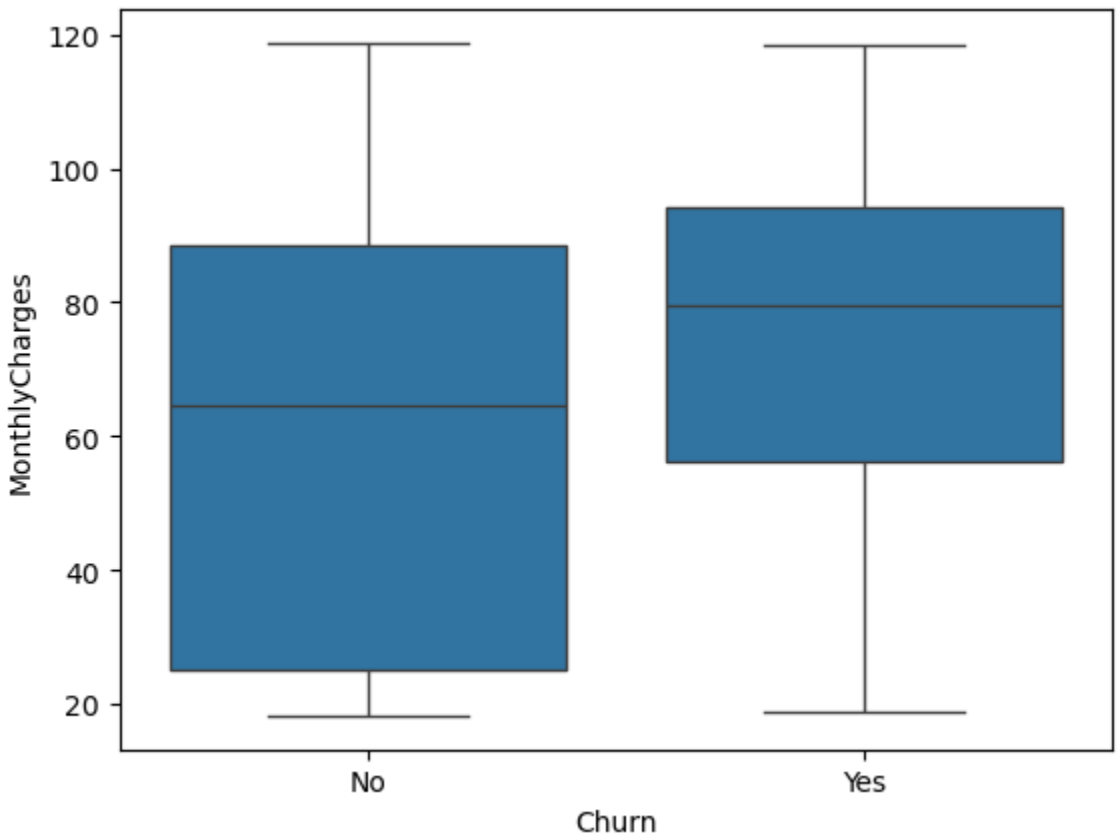
Insight

Early-stage customers are highly vulnerable.

- Action: Improve onboarding experience.

Question 3-- MonthlyCharges vs Churn

In [10]:
```
sns.boxplot(x='Churn',y='MonthlyCharges',data=df)
plt.show()
```



Insight:

Churn customers pay slightly higher monthly charges.

- Price sensitivity detected.

Revenue Loss Due to Churn

In [11]:
```
churn_revenue = df[df['Churn']=='Yes']['MonthlyCharges'].sum()
total_revenue = df['MonthlyCharges'].sum()
print(f"Churn Revenue : {churn_revenue:,}\nTotal Revenue :{total_revenue:,}"

print(f"Percentage Revenue At Risk {(churn_revenue/total_revenue).round(4)*1
```

```
Churn Revenue : 139,130.85
Total Revenue :455,661.0
Percentage Revenue At Risk 30.53%
```

Business Insight

- Although churn rate = 26.5%

- Revenue at risk = 30.5%

Churn customers generate disproportionately high revenue.

This is CRITICAL.

In [12]:
```
# Customer Lifetime Value
df['CLV']=df['MonthlyCharges']*df['tenure']
df.groupby('Churn')['CLV'].mean()
```

Out[12]:
```
Churn
No     2555.203283
Yes    1531.608828
Name: CLV, dtype: float64
```

Customer Lifetime Value Assumption

Note:

CLV estimation assumes stable monthly revenue behavior and does not include acquisition or servicing costs.

In [15]:
```
total_revenue=df['CLV'].sum()

churn_revenue = df[df['Churn']=='Yes']['CLV'].sum()

risk_percent = (churn_revenue/total_revenue)*100

print("Revenue at Risk % :",round(risk_percent,2))
```

```
Revenue at Risk % : 17.83
```

Risk Segmentation (Tenure Buckets)

In [16]:
```
df['Tenure_Group'] = pd.cut(df['tenure'],bins=[0,12,24,48,72],
                            labels=['0-1yr','1-2yr','2-4yr','4-6yr'])

pd.crosstab(df['Tenure_Group'],df['Churn'],normalize='index')*100
```

Out[16]:

Churn	No	Yes
Tenure_Group		
0-1yr	52.321839	47.678161
1-2yr	71.289062	28.710938
2-4yr	79.611041	20.388959
4-6yr	90.486824	9.513176

Insight

first 12 months = Danger Zone.

company should implement:

- welcome offers
- Loyalty onboarding programs

Pareto Customer Revenue Analysis (80/20 Rule)

In [17]:
```
#  Customer wise revenue calculate
customer_rev = (
    df.groupby("customerID")["TotalCharges"]
    .sum()
    .reset_index()
)

#  Sort highest revenue first
customer_rev = customer_rev.sort_values(
    by="TotalCharges",
    ascending=False
)

#  Cumulative Revenue
customer_rev["Cumulative_Revenue"] = (
    customer_rev["TotalCharges"].cumsum()
)

#  Total Revenue
total_revenue = customer_rev["TotalCharges"].sum()

#  Percentage
customer_rev["Cumulative_Percentage"] = (
    customer_rev["Cumulative_Revenue"] / total_revenue
) * 100

customer_rev.head()
```

Out[17]:

	customerID	TotalCharges	Cumulative_Revenue	Cumulative_Percentage
2000	2889-FPWRM	8684.80	8684.80	0.054090
5350	7569-NMZYQ	8672.45	17357.25	0.108103
6844	9739-JLPQJ	8670.10	26027.35	0.162102
6881	9788-HNGUT	8594.40	34621.75	0.215629
6264	8879-XUAHX	8564.75	43186.50	0.268971

In [18]:
```
pareto_80 = customer_rev[customer_rev['Cumulative_Percentage'] <= 80]
```

```
pareto_80.shape[0]
```

Out[18]: 2701

In [19]:
```
print(f"Total Customers : {customer_rev.shape[0]:,}")

round((pareto_80.shape[0]/customer_rev.shape[0])*100,2)
```

```
Total Customers : 7,032
```

Out[19]: 38.41

38.41% customers generate 80% revenue.

Pareto Revenue Insight

Approximately 38.41% of customers contribute to 80% of total revenue.

This indicates moderate revenue concentration rather than extreme dependency on a small customer group.

Business Implication:

• Revenue risk is distributed across a broader customer base.

• Retention programs should focus on high-value segments but overall dependency risk remains controlled.

Business Meaning:

Revenue concentration risk exists. Losing a small segment of high-value customers can significantly impact company revenue.

High-Risk Profile Extraction

In [20]:
```
high_risk = df[(df['Contract']=='Month-to-month') & (df['tenure']<12) & (df[

high_risk.shape[0]
```

Out[20]: 770

Insight

This segment = Targeted retention campaign group.

Business Action: Prioritize retention campaigns, loyalty discounts, and proactive support outreach for this segment.

STATISTICAL VALIDATION

Statistical Conclusion

Chi-square testing confirms a statistically significant relationship between contract type and churn behavior (p-value < 0.05).

This validates that churn patterns are not random and require targeted business intervention.

Is churn dependent on contract type?

In [21]:
```
from scipy.stats import chi2_contingency

cont_table = pd.crosstab(df['Contract'],df['Churn'])

chi2,p,dof,expected = chi2_contingency(cont_table)

print(f"Chi-Square : {chi2}\nP-Value :{p}")

if p < 0.05:
    print("Reject Null Hypothesis (Contract type significantly impacts churn
else:
    print("Reject to Fail Null Hypothesis")
```

```
Chi-Square : 1179.5458287339445
P-Value :7.326182186265472e-257
Reject Null Hypothesis (Contract type significantly impacts churn)
```

Ques: Do churn customers pay significantly different monthly charges?

In [22]:
```
from scipy.stats import ttest_ind

churn_yes = df[df['Churn']=='Yes']['MonthlyCharges']
churn_no = df[df['Churn']=='No']['MonthlyCharges']

t_stats,p_val = ttest_ind(churn_yes,churn_no)

print(t_stats,p_val)

if p_val < 0.05:
    print("Monthly charges significantly differ between churn and non-churn
```

```
16.47959313114872 6.760843117980302e-60
Monthly charges significantly differ between churn and non-churn customers.
```

High monthly charges increase churn probability

In [23]:
```
# T-Test - Tenuer
t_stat2,p_val2 = ttest_ind(df[df['Churn']=='Yes']['tenure'],
                           df[df['Churn']=='No']['tenure']
                          )

t_stat2,p_val2
if p_val2 < 0.05:
    print("Tenure difference is extremely significant.")
```

```
Tenure difference is extremely significant.
```

Short-tenure customers are statistically more likely to churn.

Effect size-- Cohen's d (Effect size tells practical impact)

In [24]:
```
def cohens_d(x,y):
    nx = len(x)
    ny = len(y)
    dof = nx + ny - 2
    pooled_std = np.sqrt(
    ((nx-1)*np.var(x) + (ny-1)*np.var(y)) / dof
    )
    return (np.mean(x) - np.mean(y)) / pooled_std

d_monthly = cohens_d(churn_yes,churn_no)
d_tenure = cohens_d(df[df['Churn']=='Yes']['tenure'],
                    df[df['Churn']=='No']['tenure']
                        )
print(d_monthly,d_tenure)
```

```
0.4449241885346609 -0.8569679897884734
```

Statistical Validation Summary

- Contract type significantly affects churn (Chi-square test).
- Monthly charges differ significantly between churn and non-churn customers (T-test).
- Tenure has a strong statistical impact on churn.
- Tenure shows large effect size, meaning it has strong practical importance.

Conclusion: Churn is not random. It is driven by identifiable and statistically significant factors.

Executive Recommendations

1. Early tenure customers require proactive onboarding programs.
2. Month-to-month contract customers represent highest churn risk and should be incentivized toward longer commitments.
3. Pricing sensitivity among higher monthly charge customers requires bundled value strategies.
4. Approximately 38% of customers contribute to 80% of lifetime value emphasizing priority retention segmentation.

Final Business Recommendations

1. Offer loyalty discounts for month-to-month customers.
2. Improve onboarding experience for first 12-month customers.
3. Monitor high monthly charge segments for early churn signals.
4. Prioritize retention campaigns for Pareto high-value customers.

Expected Business Impact:

Reducing churn among high-value customers can significantly improve revenue stability and customer lifetime profitability.

Tools Used

- Python — Pandas, NumPy
- Visualization — Matplotlib, Seaborn
- Statistical Testing — SciPy
- Business Analytics — Pareto Analysis, CLV Modeling